

# **Assignment 2 – Scenario 1**

*Breast cancer dataset*

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## 1. Introduction

This project aims to create a model to predict cancer detection using machine learning methods. After some evaluation in different ways, it will employ Random Forest Classifiers and Logistic Regression to provide accurate diagnoses for patients and then suggest the most appropriate model.

## 2. Data understanding and preprocessing

Using pandas, import the dataset with 569 rows and six columns characterising cancer. The 'diagnosis' column is the dependent variable, representing a binary value (0 for malignant and 1 for benign). The remaining columns are independent variables providing continuous figures for tumor characteristics, which help to analysis the diagnosis result is malignant or benign.

Throughout the dataset, there is no missing and duplicated values. After that, we grouped by the diagnosis result and calculated the mean values for each tumour characteristic (**Figure 1**). This can help us define the average value of each factor impacting the malignant or benign diagnosis that may affect to the result of predictions.

```
df.groupby('diagnosis').mean()
```

|           | mean_radius | mean_texture | mean_perimeter | mean_area  | mean_smoothness |
|-----------|-------------|--------------|----------------|------------|-----------------|
| diagnosis |             |              |                |            |                 |
| 0         | 17.462830   | 21.604906    | 115.365377     | 978.376415 | 0.102898        |
| 1         | 12.146524   | 17.914762    | 78.075406      | 462.790196 | 0.092478        |

**Figure 1**

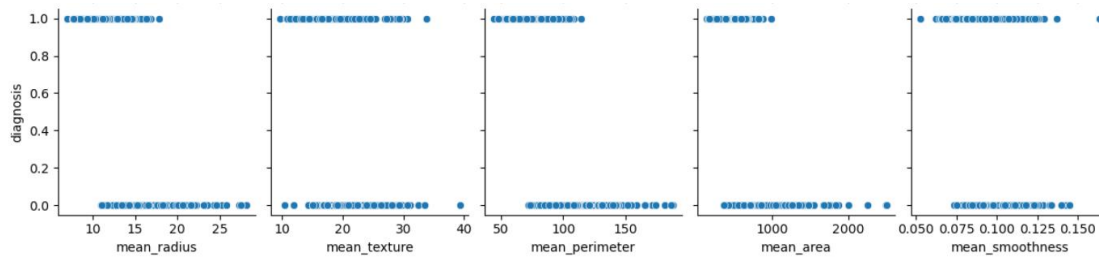
## 3. Model Selection

Upon examining the correlation matrix, **as shown in Figure 2**, we found extremely strong linear relationships between mean\_radius, mean\_perimeter, mean\_area, and diagnosis. This indicates that these factors are crucial in identifying the independent variable. However, in mean\_texture and mean\_smoothness we observed weak linear relationships, suggesting that these characteristics may not significantly impact the final diagnosis.

|                 | mean_radius | mean_texture | mean_perimeter | mean_area | mean_smoothness | diagnosis |
|-----------------|-------------|--------------|----------------|-----------|-----------------|-----------|
| mean_radius     | 1.000000    | 0.323782     | 0.997855       | 0.987357  | 0.170581        | -0.730029 |
| mean_texture    | 0.323782    | 1.000000     | 0.329533       | 0.321086  | -0.023389       | -0.415185 |
| mean_perimeter  | 0.997855    | 0.329533     | 1.000000       | 0.986507  | 0.207278        | -0.742636 |
| mean_area       | 0.987357    | 0.321086     | 0.986507       | 1.000000  | 0.177028        | -0.708984 |
| mean_smoothness | 0.170581    | -0.023389    | 0.207278       | 0.177028  | 1.000000        | -0.358560 |
| diagnosis       | -0.730029   | -0.415185    | -0.742636      | -0.708984 | -0.358560       | 1.000000  |

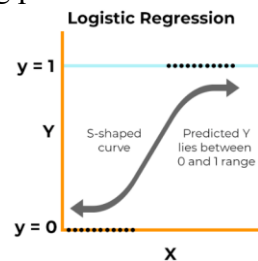
**Figure 2**

From the plots as seen in **Figure 3**, it is also apparent that the distribution and relationship of diagnosis (0 for malignant and 1 for benign), the data for malignant (0) is significantly higher than that for benign (1) in the three linear variables. Besides that, there is a slightly imbalanced class. For the aforementioned reasons, Random Forest and Logistic Regression would be the most appropriate for this project.



**Figure 3**

Logistic regression is a method used for binary classification problems. It uses the logistic function (**Figure 4**) to model the linear relationship between independent variables and the probability of a binary outcome. The coefficients indicate the strength and direction of the relationship between each feature and the outcome. This method provides a probabilistic framework for making predictions.

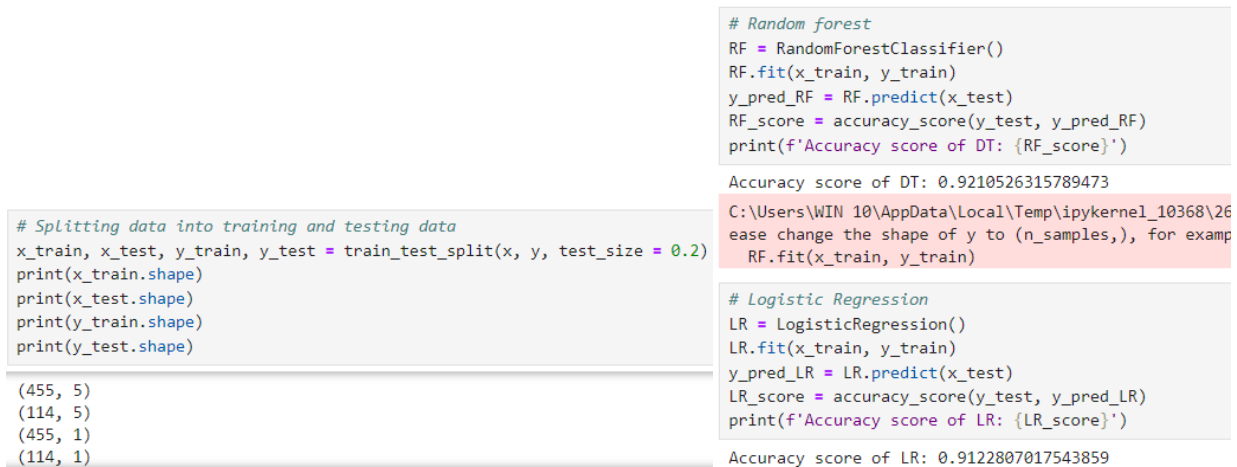


**Figure 4**

The Random Forest Classifier is an ensemble learning technique used for classification problems. It combines multiple decision trees, which solve the imbalanced class problems, to make final predictions in linear and non-linear relationships. Each decision tree is created using a different subset of the data and random subsets of features. The predictions from all the trees are then aggregated, typically using majority voting for classification tasks. Therefore, each tree in the forest is unique, providing its prediction.

## 4. Model Performance Comparison

The data was split into 80% for training and 20% for testing, a standard ratio for machine learning models. The accuracy scores are 92.1% for Random Forest Classifier and 91.2% for Logistic Regression. (**Figure 5**)



**Figure 5**

By adjusting the training and testing ratio to 70% and 30% respectively, we were able to gather a more diverse set of data points for evaluation, which in turn allowed for more accurate predictions. Training with 70% of the data still proved to be sufficient without leading to underfitting or overfitting issues. Additionally, including a `random_state` parameter of 42 ensured stability in the splitting of training and testing sets. Moreover, setting the `n_estimators` parameter, to 250. As a result, we achieved 97% accuracy for Random Forest Classifiers and 93.5% for Logistic Regression (**Figure 6**). As Random Forest has a better ability to handle imbalanced data with linear or non-linear relationship by applying the multiple decision trees concept.



**Figure 6**

## 5. Testing and Model Demonstration

To validate the model, first, we take a set of independent variables from the dataset, leading to a malignant diagnosis. Then, we carry out the testing in two models, resulting the expected results. However, we record two different outcomes when we keep the mean\_texture and mean\_smoothness values. (**Figure 7**)

```
# Random forest
RF_pred_1 = RF.predict([[15.78, 22.91, 105.7, 782.6, 0.1155]])
print(RF_pred_1)
RF_pred_2 = RF.predict([[13.14, 22.91, 70.6, 500.8, 0.1155]])
print(RF_pred_2)

# Logistic Regression
LR_pred_1 = LR.predict([[15.78, 22.91, 105.7, 782.6, 0.1155]])
print(LR_pred_1)
LR_pred_2 = LR.predict([[13.14, 22.91, 70.6, 500.8, 0.1155]])
print(LR_pred_2)
```

[0]  
[0]

[0]  
[1]

**Figure 7**

This is because Logistic Regression applies a single linear decision boundary, leading to potential misclassification in weak linear relationships with outcomes, while Random Forest can adapt decision boundaries for accurate predictions in complex interaction dataset.

## 6. Recommendation

The results indicate that the Random Forest Classifier performs better than the other in managing non-linear relationships and imbalanced data. It is suggested for this project after a thorough evaluation of both models' model scores and testing outcomes. It is the better choice for cancer detection because of its capacity to combine several decision trees and simulate intricate relationships, which yields more accurate forecasts. From **Figure 8**, we also concluded that the predictions from RandomForest indicate a higher evaluation compared to Logistic Regression.

```
RF_report = classification_report(y_test, y_pred_RF)
print(f'Classification report of RandomForest: {RF_report}')
```

| Accuracy score of RandomForest: |      | precision | recall | f1-score | support |
|---------------------------------|------|-----------|--------|----------|---------|
| 0                               | 0.94 | 0.98      | 0.96   | 0.96     | 63      |
| 1                               | 0.99 | 0.96      | 0.98   | 0.98     | 108     |
| accuracy                        |      |           | 0.97   |          | 171     |
| macro avg                       | 0.96 | 0.97      | 0.97   |          | 171     |
| weighted avg                    | 0.97 | 0.97      | 0.97   |          | 171     |

```
LR_report = classification_report(y_test, y_pred_LR)
print(f'Classification report of LogisticRegression: {LR_report}')
```

| Accuracy score of RandomForest: |      | precision | recall | f1-score | support |
|---------------------------------|------|-----------|--------|----------|---------|
| 0                               | 0.89 | 0.94      | 0.91   | 0.91     | 63      |
| 1                               | 0.96 | 0.94      | 0.95   | 0.95     | 108     |
| accuracy                        |      |           | 0.94   |          | 171     |
| macro avg                       | 0.93 | 0.94      | 0.93   |          | 171     |
| weighted avg                    | 0.94 | 0.94      | 0.94   |          | 171     |

**Figure 8**